Miles V. Vollmer

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PROFESSIONAL SUMMARY

AI and performance engineer with 3+ years of professional experience. I build, integrate, and optimize enterprise systems, and my results are on the record: $2M+ per month in infrastructure savings, an award-winning enterprise dashboard, single-handed AI-accelerated platform migrations, and peer-reviewed machine learning research. Trained in Northeastern's CS/AI program, I work in any language or stack and deliver improvements at every scale, from a single query to an enterprise platform.

EDUCATION & CERTIFICATIONS

B.S. in Computer Science with AI Concentration, Northeastern University, Boston, MA (2024)

Certifications: AWS Cloud Practitioner, BMC Certified Associate, BMC AIOps

TECHNICAL SKILLS

Engineering: Full-Stack Development, System Integration, ETL Pipelines, RESTful APIs, Performance & Cost Optimization, Cloud Capacity Planning, Observability, CI/CD

Languages: Python, Java, SQL, TypeScript, JavaScript, R, Bash, HTML/CSS (language-agnostic; productive in any stack)

Cloud & Data Platforms: Databricks (AI/BI, Unity Catalog), AWS (EC2, S3), Google Cloud Run, Grafana, Tableau, Power BI, dbt, Docker, Kubernetes, Git, Dynatrace, BMC Helix, NetSuite

AI & Machine Learning: LLM Agents & RAG, Google ADK, AI-Assisted Development (Claude Code, context engineering), PyTorch, TensorFlow, HuggingFace, NLP, Reinforcement Learning, Ensemble ML & SHAP Pipelines (published)

PROFESSIONAL EXPERIENCE

Performance Engineering Consultant – Moviri *Jan 2025 – Present*

Consulting engineer to global banking and telecommunications clients. Engagements below span cost optimization, full-stack development, platform migration, and AI-assisted automation.

Senior Consultant, Major UK Bank (Cloud Migration & Infrastructure Optimization):
Led capacity analysis across ~10,000 overprovisioned servers; the resulting rightsizing cut infrastructure spend by **$2M+ per month**. Built the front and back end of a Business Units drill-down dashboard, winner of the **Innovation Award at the BMC Helix London Roadshow**. Designed a three-tier, bulk-cached metadata resolution chain for 2,300+ hosts, eliminating per-host API calls. Built a Datamart Dependencies Extractor (graph cycle detection, memoization), now reusable company IP. Co-authored ~50 pages of the contractual Construction Document.

Major Italian Telecommunications Provider (BI Platform Modernization):
Sole engineer re-platforming **450 Grafana dashboards (7,000+ visualizations)** into Databricks on a fixed timeline: ~20 dashboards and ~335 visualizations per working day, before verification and rework. Architected the AI-assisted, 15-tool toolchain behind this pace (context curation, decision log, 63-test regression suite). Absorbed scope growth, including 150+ source dashboards modified after export, while improving on original functionality. Built the AWS cost model ($89K/yr); root-caused a permissions blocker on 87% of data sources.

Internal AI & Operational Automation (Moviri):
Designed and piloted HOAO, a multi-agent framework in which LLM agents write, sandbox-test, and validate their own tools under human approval. Built a full-stack meal voucher application and a NetSuite-integrated onboarding agent. Automated monthly product releases via YAML CI/CD.

Research Assistant – Evolutionary Genomics Lab, Northeastern University *May 2021 – Sep 2021, Sep 2022 – Aug 2024*

Built ensemble machine learning and SHAP pipelines (Random Forest, SVM, MLP) on 16S rDNA sequencing data; **97%+ classification accuracy**. Identified the taxa driving disease; co-authored a peer-reviewed paper in *Environmental Microbiology*.

Data Research Analyst – Iafrate Lab, Massachusetts General Hospital (MGH) *Jan 2022 – Aug 2022*

Built automated Genome-Wide Association (GWA) workflows for COVID-19 genetic data; demographic visualizations informed clinical policy discussions.

PUBLICATIONS

Selwyn, JD, Despard, BA, Vollmer, MV, Trytten EC, Vollmer, SV (2024). "Machine Learning based identification of putative coral pathogens in endangered Caribbean staghorn coral." *Environmental Microbiology* 26 (9): e16700. Link to publication

SELECTED PROJECTS

Distributed Genomic Pipeline (AWS/Snakemake): 290-genome Snakemake pipeline on fault-tolerant EC2 spot instances with S3 transfer optimization; ~70% lower compute cost.

Game-Playing AI (paper)**:** ReBeL-style self-play and CFR/MCCFR agents for 6-player No-Limit Hold'em; outperformed Q-learning baselines. Minimax 2v2 chess engine with alpha-beta pruning.